BRETT D. VINCENZINI

RESEARCH CHEMISTRY | APPLIED & FUNDAMENTAL SCIENCE

Philadelphia, PA • brett.vincenzini@gmail.com • (916) 899-3722 • LinkedIn

PROFESSIONAL PROFILE

Ph.D. inorganic chemist spanning fundamental molecular research and applied chemical-process development. Synthesized novel organometallic complexes, developed novel experimental & analytical methods, elucidated electronic structure–reactivity relationships, and designed an acid-free process for nickel and cobalt recovery from laterite ores. First-author publications in *Nature Chemistry*, *Journal of the American Chemical Society*, and *Coordination Chemistry Reviews*.

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow | University of Pennsylvania ***Dec 2025 – Present***

- Developed and optimized an integrated, acid-free **hydrometallurgical process** for recovering nickel and cobalt from mineralogically diverse laterite feedstocks.
- Evaluated process compatibility with 10 distinct ore bodies; achieved average Ni and Co recoveries of 80% and 55%, respectively, under optimized single-pass extraction conditions.
- Established a selective precipitation method to produce Ni- and Co-enriched products, including a Ni-rich product exceeding **70% Ni purity** on a metals basis.
- Integrated spectroscopic (**Raman**, Fe K-edge X-ray Absorption Spectroscopy (**XAS**)), mineralogical (**PXRD**), and **porosity** characterization with quantitative data analysis to establish relationships between thermal pretreatment, chemical speciation, material properties, and metal extraction performance.
- Co-authored a university **invention disclosure** based on the end-to-end hydrometallurgical processing technology.

Graduate Research Fellow | University of Pennsylvania ***Jan 2021 – Dec 2025***

- Defined the scope and objectives of 4 research projects by identifying knowledge gaps and salient research questions through systematic literature surveys.
- Designed and continuously updated experimental roadmaps for each project to pursue research questions to satisfactory conclusions.
- Organized and managed **interdisciplinary collaboration** on 4 research projects involving contributors from national laboratories and academic institutions.
- Developed/optimized syntheses for **9 unprecedented organometallic complexes** including 3 comprising direct Ce(IV)–C bonds (2 Hf, 1 Zr, 2 Th, and 1 U complexes).
- Integrated solid-state reactivity, spectroscopic, and theoretical studies to demonstrate the importance of 4f-orbital covalency in determining chemical reactivity patterns in lanthanide compounds for the first time.
- Developed **SC-XRD methods** to monitor and characterize thermal and photochemical solid-state reactions occurring in single-crystals and to trap and study otherwise unstable reaction intermediates and products.
- Built and validated a **mathematical model in Python** to describe light-driven reactivity within single-crystals, enabling the extraction of quantitative kinetic and thermodynamic information. Model improved the precision of the calculated kinetic isotope effect by 51%.
- Applied solution-phase spectroscopic techniques to elucidate **reaction mechanisms** and decomposition pathways in an organocerium system.

- Discovered a new class of Ce(IV) metallocenes exhibiting strongly correlated electronic states; leveraged Ce L_{III}-edge XAS measurement and modeling capabilities to investigate the structural and symmetry-dependent origins of these properties.

Undergraduate Researcher | San Diego State University ***Jan 2017 – May 2020***

- Synthesized and tested two ruthenium-based electrocatalysts for water oxidation in the context of clean-energy catalysis.

EDUCATION

University of Pennsylvania | Ph.D. in Inorganic Chemistry ***Dec 2025***

San Diego State University | B.S. in Chemistry ***May 2020***

SELECTED PUBLICATIONS | Google Scholar

- **Vincenzini, B. D.**, Yu, X., Paloc, S., Smith, P. W., Gupta, H., Pandey, P., Kent, G. T., Ordonez, O., Keller, T., Gau, A. M., Bacon, P. J., Minasian, S. G., Hayton, T. W., Autschbach, J., Schelter, E. J., "4f-orbital Covalency Enables a Single-Crystal-to-Single-Crystal Ring-Opening Isomerization in a Ce^{IV}-Cyclopropenyl Complex" *Nat. Chem.*, **2025**. DOI: 10.1038/s41557-025-01791-2.
  Peplow, M., "**Lanthanide orbitals spring a reaction surprise**", Chemical & Engineering News, 2025.
- **Vincenzini, B. D.**, Schelter, E. J., "Cerium(IV) Compounds Comprising Ce–C, Ce=N, Ce=O Bonds" *Coord. Chem. Rev.*, **2025**, 544, 216968.
- **Vincenzini, B. D.**, Yu, X., Gomez, A., Gau, M. R., Autschbach, J., Schelter, E. J., "Mechanistic Study of a Thermal- and Light-Driven Single-Crystal-to-Single-Crystal Transformation of an Organocerium Complex" *J. Am. Chem. Soc.*, **2026**. DOI: 10.1021/jacs.6c14543
- **Vincenzini, B. D.**, Sergentu, D. C., Zaichenko, A., Booth, C. W., Smith, P. W., Gupta, H., Gau, M. R., Autschbach, J., Minasian, S. G., Schelter, E. J., "Strong Electron Correlations and Multi-Configurational Ground States of Cp_3CeX Compounds Manifest through Metal-Ligand Phi-Bonding Interactions" *Chem. Sci.* (**Submitted**).

TEACHING & MENTORSHIP

Teaching Assistant | University of Pennsylvania ***Sep 2020 – May 2021***

- Prepared weekly slide decks from lecture materials for introductory and advanced organic chemistry laboratory courses. Led weekly laboratory sessions and held office hours to answer student questions.

Undergraduate & Graduate Research Mentor | Schelter Laboratory

- Mentored two undergraduate and one junior graduate student in relevant scientific concepts and foundational synthetic and analytical techniques.

TECHNICAL SKILLS

Experimental: Inorganic, organic, & organometallic synthesis; Schlenk techniques; air-free solvent systems; chromatography; inert atmosphere glovebox use/maintenance; high energy ball milling; tube & muffle furnaces.
Instrumentation: SC-XRD; XAS/XANES; NMR (1H, 13C, 31P, 19F, 7Li; 2D/VT); Raman; IR; UV–vis; fluorescence; mass spectrometry; HPLC; GC; potentiostat; BET adsorption; particle size analysis; ICP-OES
Software: Python (mathematical modeling); ChemDraw; Mnova; Olex2; Mercury; TopSpin; Athena; Fityk; Adobe Illustrator

HONORS & AWARDS

Excellence in Chemistry Graduate Research Award | University of Pennsylvania (2024)

Graduate Committee's List Award | University of Pennsylvania (2021)